Matrix Rounding Problem

This application is concerned with consistent rounding of the elements, row sums, and column sums of a matrix. We are given a 3×3 matrix of real numbers x_{ij} with row sums vector r and column sums vector c. We can round any real number $a \in \{x_{11} \ldots x_{33}, r_1 \ldots r_3, c_1 \ldots c_3\}$ to the next smaller integer $\lfloor a \rfloor$ or to the next larger integer $\lceil a \rceil$, and the decision is completely up to us. The problem requires that we round all matrix elements and the following constraints hold:

- the sum of the rounded elements in each row is equal to the rounded row sum r_i
- the sum of the rounded elements in each column is equal to the rounded column sum c_i

The objective is to maximize the sum of rounded matrix entries.

a) Formulate as the maximum flow problem.

b) Are the resulting flows integers? Why yes/no?

c) Is the problem feasible for any admissible input?

Production Planning

We remind you that your formulation must be linear, i.e., no multiplication of variables, no implications, no logical expressions, etc. All non-linearities you might require, need to be linearized so that they can be written as linear expressions.

A company plans to produce five products. Formulate as an ILP problem. Given the following constraints, find optimal volume of each product while maximizing the profit:

1. p_i is the number of hours needed for one piece of product i
2. P is the company labor capacity in hours
3. the volume of each product is at most 1000 pieces
4. the volume of product 1 is an even number
5. at least one of the following constraints holds:
   - the volume of product 2 plus product 3 is at least 100
   - the volume of product 4 plus product 5 is at most 200
6. the income generated by one piece of product i is given by v_i
7. fixed costs (i.e. expenses that are the same for 1 piece as well as for 1000 pieces) needed to hire the machines for product i are given by f_i
8. labor cost for one piece of product i is given by c_i
9. profit is given as the sum of the incomes minus the sum of the labor and fixed costs

Longest paths

Using Floyd's algorithm calculate square matrix q, where q_{ij} is the length of the **longest path** from vertex i to vertex j (you do not need to calculate matrix of predecessors p). Indicate the steps of the algorithm (values of matrix l for 4 iterations of the main loop).

